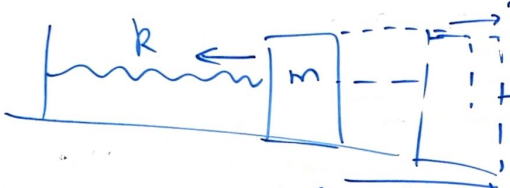
15.7.24

Waves Lec 14

Driven and damped osc.



$F_f = -b\dot{x}$

$F_{sp} = -kx$

$m\ddot{x} = -kx - b\dot{x} + F_0 \cos\omega t$

$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = f_0 \cos\omega t$

$\gamma = \frac{b}{m}, \quad \omega_0^2 = \frac{k}{m}, \quad f_0 = \frac{F_0}{m}$

$z = x + iy, \quad \tilde{F} = F_0 e^{i\omega t}$

$\ddot{z} + \gamma\dot{z} + \omega_0^2 z = F_0 e^{i\omega t}$

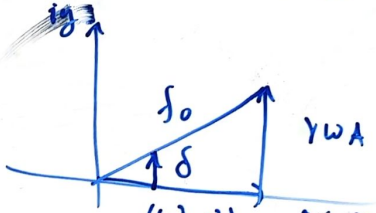
$x = Re[z]$

$z = A e^{i(\omega t - \delta)}$

$\dot{z} = i\omega A e^{i(\omega t - \delta)}$

$\ddot{z} = -\omega^2 A e^{i(\omega t - \delta)}$

$(\omega_0^2 - \omega^2) A + i\gamma\omega A = f_0 e^{i\delta} = f_0[\cos\delta + i\sin\delta]$



$\omega < \omega_0$

$A(\omega_0) = \frac{f_0}{\gamma\omega_0}$

$A(0) = f_0/\omega_0^2$

$D(\omega) = \left[(\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2\right]$

$A(\omega) = \frac{f_0}{\sqrt{D(\omega)}}$

$A'(\omega) = 0 \Rightarrow$

$D'(\omega) = 0 = D'(\omega_m)$

$\omega_m^2 = \omega_0^2 - \frac{\gamma^2}{2}$

$A(\omega) = \frac{f_0}{\left[(\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2\right]^{1/2}}$

$\delta(\omega) = \tan^{-1}\left[\frac{\gamma\omega}{\omega_0^2 - \omega^2}\right]$

$A(\omega)$ is max. at $\omega_m^2 = \omega_0^2 - \frac{\gamma^2}{2}$

damped natural freq. $\omega_d^2 = \omega_0^2 - \gamma^2/4$

$$D(\omega) = (\omega - \omega_m)^2 + \gamma^2 \omega_d^2$$

$$D(\omega_m) = \gamma^2 \omega_d^2$$

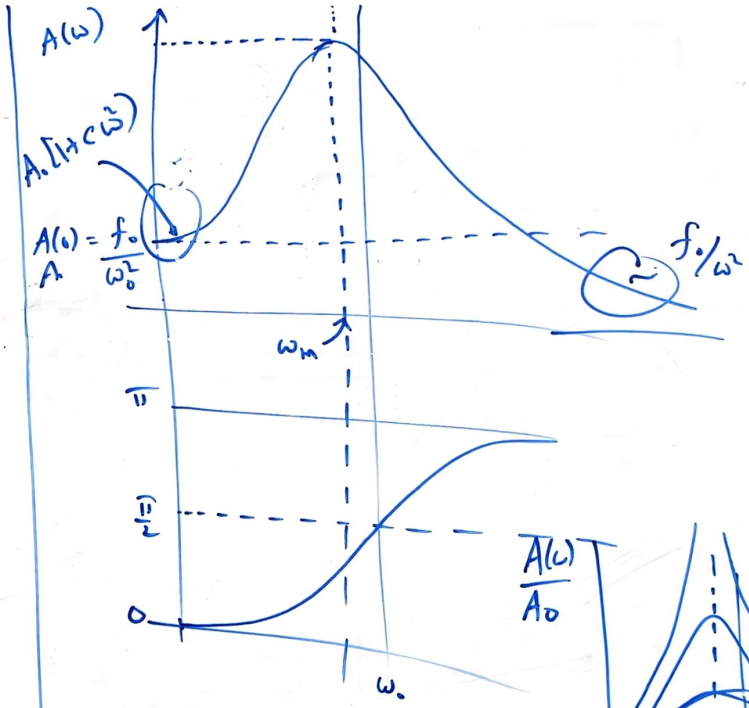
$$A_m = A(\omega_m) = \frac{f_0}{\sqrt{D(\omega_m)}} = \frac{f_0}{\gamma \omega_d}$$

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2/4} = \omega_0 \left[1 - \gamma^2/4\omega_0^2\right]^{1/2}$$

Q = "quality factor" $= \frac{\omega_0}{\gamma}$

$$\omega_d = \omega_0 \left[1 - 1/4Q^2\right]^{1/2}$$

$$\omega_m = \omega_0 \left[1 - 1/2Q^2\right]^{1/2}$$



$$\frac{\omega_0}{\gamma} = Q \Rightarrow \gamma = \omega_0/Q$$

$$\frac{A(\omega_m)}{A(0)} = \frac{A_m}{A_0} = \frac{f_0/\gamma\omega_d}{f_0/\omega_0^2}$$

$$= \frac{\omega_0^2}{\gamma \omega_d} = \frac{\omega_0/\gamma \cdot \omega_0}{\omega_0\left[1 - 1/4Q^2\right]^{1/2}}$$

$$= \frac{Q}{\left[1 - 1/4Q^2\right]^{1/2}}$$

$$\delta(\omega) = \tan^{-1}\left[\frac{\gamma\omega}{\omega_0^2\left[1 - \omega^2/\omega_0^2\right]}\right]$$

$$= \tan^{-1}\left[\frac{\omega/\omega_0}{Q\left(1 - \omega^2/\omega_0^2\right)}\right]$$

